

# Scripting System for Ember LIMS

Design and Implementation of a secure C# Scripting  
Layer in a .NET ASP Application

# Project Context



- Migration from LX to Ember
- Scripting System for Ember LIMS
- LIMS = Laboratory Information Management System



# What is the scripting system?

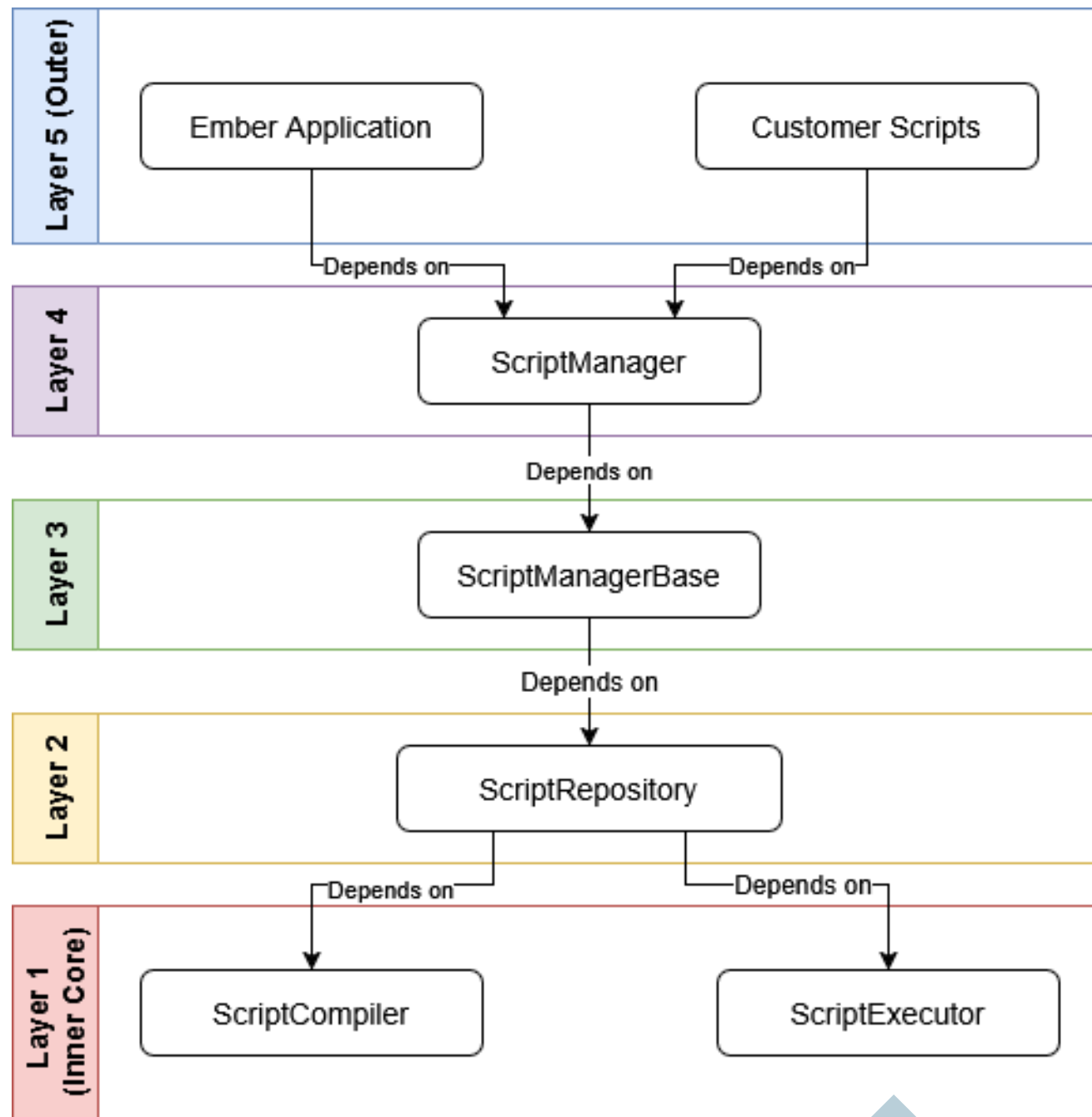
- Customer customization
- Without changes to the source-code of Ember
- Script = small piece of code

# Previous scripting system

- Previous system based on Python
- Lacked strong validation and security controls

# Goals of the new system

- Written in C#
- Security
- Backwards-compatibility
- Simplicity and usability



# Compilation and execution model

- JIT vs AOT
- Ahead-of-time compilation using Roslyn
- Compiled scripts stored as byte array in PostgreSQL
- Timeout handling using cancellation token

# Validations performed in the Compiler

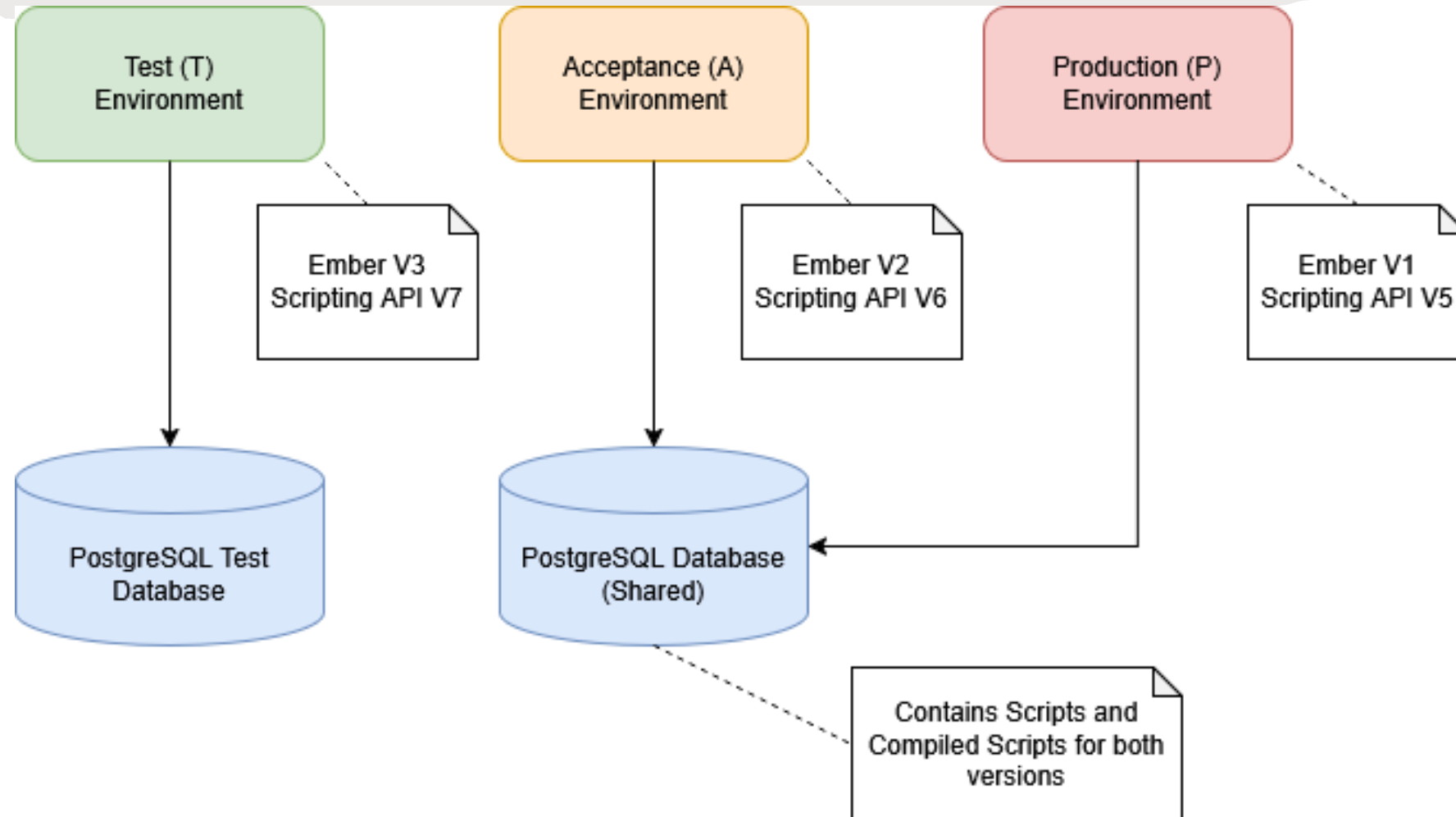
- Validations performed before insertion into DB
- Example: Namespace restrictions for dangerous APIs



# Versioning system

Allowing for safe backwards compatibility

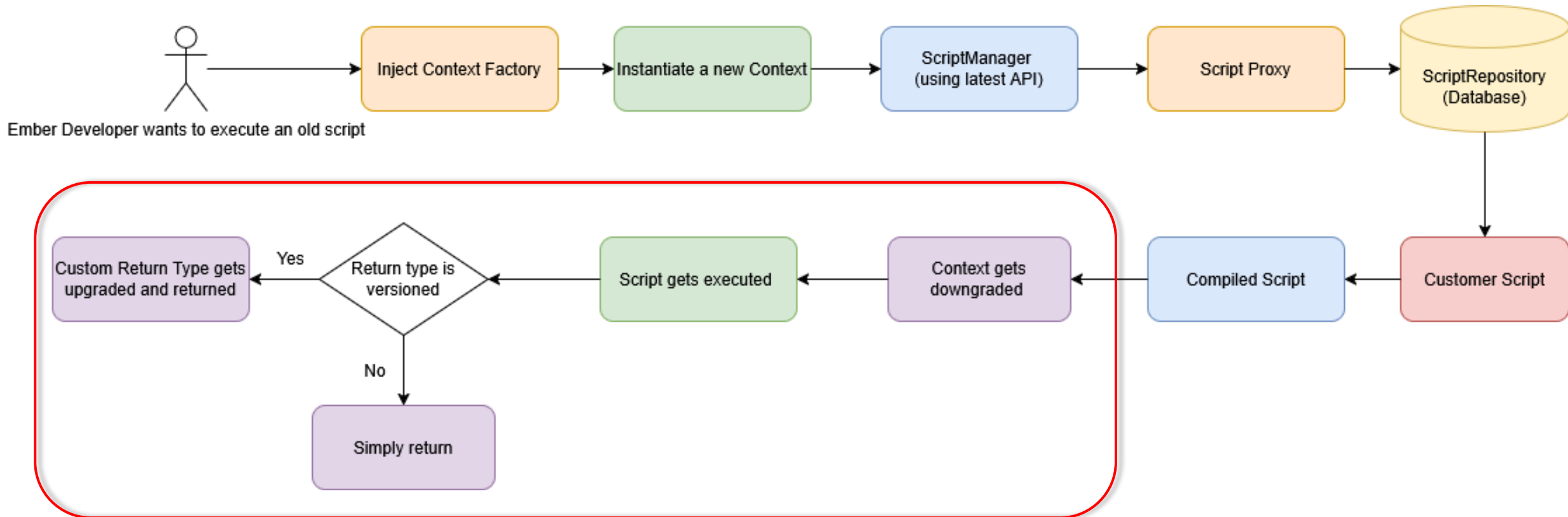
# Multiple environments share one DB



# Versioning

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- Script interfaces are versioned
- Contexts are versioned
- Custom return types are versioned



# Execution of a script written for an older version

# Upgrade/Downgrade strategy

- Developer interacts only with the latest versions
- New contexts are downgraded for old scripts
- Old return values are upgraded to recent types
- Simplifies the usage

# Evaluation



- Test-Driven Development
- Many Integration tests
- Standard operations covered in tests

# Limitations and future work

- Loop analysis can still be circumvented
- Instead of a blacklist switch to whitelist
- Improve memory handling in the executor
- Add custom Roslyn analyzers and better tooling
- Addition of a high-speed cache
- Too few unit-tests

# Conclusion

- Successful proof-of-concept
- Proves feasibility of a scripting system written in C#
- That is secure, backwards-compatible and fast
- Balances the stability of the system
- With the power customer scripts get
- Not production ready





**THANK YOU**

